MIL-S-36471(DSA-DM)
14 March 1966

MILITARY SPECIFICATION

DISK, COLOR STANDARDS,
ANTI-ICING ADDITIVE

1. SCOPE

1.1 This specification covers a color standard disk used in determining the concentration of anti-icing additive in JP-4 fuel.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up, Paperboard.

Military

MIL-C-00538	Comparator, Color, Hydrogen Ion and Residual Chlorine.
MIL-C-36255	Cells, Colorimeter.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6630

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STANDARDS

Federal

Fed. Test Method Std. 791 Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.1), prior to the start of production of the contract quantity, one preproduction sample of the color standard disk shall be assembled, completely finished, and tested by the contractor. A Master Standard disk for reference use is available from the contracting officer. The sample shall be submitted, together with the contractor's inspection and test reports, to the contracting officer for examination and testing to determine compliance with the requirements of this specification (see 3.4, 3.4.1 and 3.4.2). Written approval of the materials, construction, and performance of the sample shall be obtained from the contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Further production of the item by the contractor, prior to this approval, shall be at the contractor's risk. Accepted preproduction sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the preproduction sample is not approved, the contractor shall submit a new sample for approval, or modify the existent sample so as to remove the deficiencies cited.

3.2 Material.

3.2.1 Plastic. Disks shall be made of black, opaque, molded ABS plastic material (acrylonitrile, butadiene and styrene monomers) or phenolic type thermosetting resin. Plastic shall be free of warp, twist, cracks, chipped edges, surface blisters, uneven surfaces, scratches, dents, heat marks, fins, or burrs. Buffing or polishing may be employed, if necessary.

3.2.2 Glass color standards. Glass for color standards, on the disks, shall be of the highest quality and shall be permanent, nonfading, and uniform in color. Glass standards shall be free of cracks, scratches, seeds, blisters, inclusions, and other defects that may adversely affect the optical characteristics. Glass standards shall show no cracking, chipping or other defects. If glass standards are made up of laminated material, there shall be no delamination, when tested as specified in 4.4.2.2.

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3.3 Style and design. Disk shall consist of a flat circular plastic plate containing eight calibrated color glass standards with identifying percentage values on circular plastic inserts and shall be in accordance with figure 1. Color glass standards shall be 9/16 ± 1/32 inch in diameter and shall be of a non-fading, permanent color. The percentage values shall be permanently marked in black print on securely affixed, off-white color, plastic inserts and shall be located at an angle of 140° ± 3° from the corresponding color standards when measured in a counter-clockwise direction. Values on the disks shall be so positioned as to be upright when viewed through the reading port of the comparator. The inserts shall be flush with the flat surface of the disk. The color glass standards shall be suitable for use in diffused daylight or artificial light. The glass standards shall be secured by means of phosphor-bronze, spring retainer rings with a black Ebonal finish, so that they shall not fall out during use and still be replaceable if necessary.

3.4 Disk characteristics. Color standard disks shall be suitable for determining the percent (by volume) of anti-icing additive in aviation fuel and shall be calibrated to measure the following values: 0.00, 0.04, 0.06, 0.08, 0.10, 0.12, 0.14, and 0.16 percent by volume. The color values of the color glass standards in disks shall be made and compared to liquid color standards prepared as specified in 4.4.1.1. Tolerances shall be as specified in 3.4.1.

3.4.1 Tolerances. Percentage values (by volume) of the color glass standards shall be accurate within the tolerances indicated below, when compared with the liquid standard solutions specified in 4.4.1.1.

Percent Value	Tolerance
0.00	-0.00 +0.01
0.04	-0.01 +0.005
0.06	±0.005
0.08	±0.005
0.10	±0.005
0.12	±0.005
0.14	±0.005
0.16	±0.005

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3.4.2 Accuracy for acceptance of disks. Color glass standards of each production disk shall match the corresponding color standards of the pre-production sample disk when tested as specified in 4.4.2.1

3.5 Finish. Disks shall have a black, smooth finish (except for inserts and identification marking labels) and shall have no rough or sharp edges. Disks shall show no unsightly finish caused by chipping, filing or grinding of ridges.

3.6 Identification marking. Disks shall be permanently marked in black print on a securely affixed, off-white color, rigid plastic label as follows:

Anti-Icing
26 MM CELL
Percent by Volume
Name or registered trademark of the manufacturer.

Labels shall be even with the flat surfaces of the disk.

3.7 Workmanship. Workmanship shall be first class throughout. Disks shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties)of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

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6. NOTES

6.1 Instructions for preproduction sample approval. The contractor shall inform the contracting officer through the Government inspector when the preproduction samples are ready for examination and testing. The contracting officer shall inform the contractor when and where the samples are to be sent (see 3.1).

6.2 Subject disk is used with the following items:

a. FSN 6630-023-2297 Comparator, Color referencing Military Specification MIL-C-00538 and applicable portions of DPSC Drawing 20651.

b. FSN 6630-418-3580 Cell Colorimeter, 13 by 26 by 77 mm, referencing Military Specification MIL-C-36255.

6.3 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.2).

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

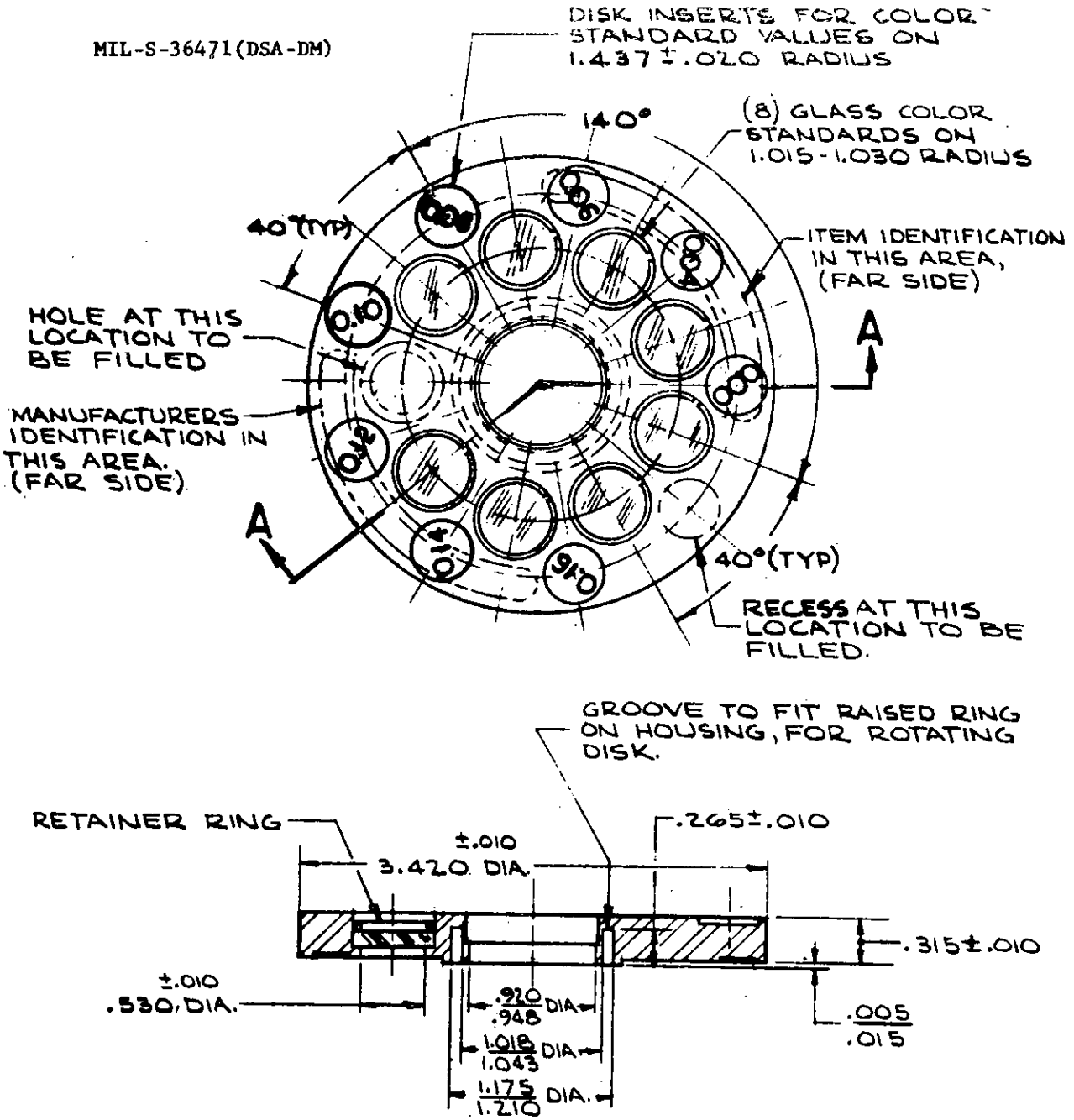
Custodians:
Army - MU
Navy - SH
Air Force - 67

Review activities:
Army - MU
Navy - SH
Air Force - 67

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SECTION A-A

FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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